# A BURGERS–FISHER EQUATION FOR PRIME GAPS: SPECTRAL PROOF OF CONVERGENCE TO POISSON

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Abstract. We prove that the velocity field $u = -\partial_v \log f$, where $f$ is the probability density of normalized prime gaps $v = g/\log x$, satisfies a Burgers–Fisher equation in a porous medium:

$$\partial_t u + 2v\, u\, \partial_v u = v\, \partial_v^2 u + (v{+}2)\, \partial_v u + u(1{-}u).$$

The Fisher–KPP reaction term $u(1-u)$ emerges naturally from the Fokker–Planck structure and drives $u$ toward 1, which corresponds to the exponential (Poisson) distribution. The linearized perturbation operator has associated Laguerre polynomial eigenfunctions with spectral gap 1. Under a decaying sieve forcing, perturbations vanish as $O(1/(\log x)^{\beta})$. Numerical verification at 11 scales from $10^3$ to $10^8$ shows the mutual information between consecutive gaps decays as $(\log x)^{-0.19}$ ($R^2 = 0.986$) and that prime $p = 3$ carries 94% of the sieve signal.

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## 1. Introduction

Gallagher [1] proved, conditionally on the Hardy–Littlewood conjecture [2], that prime counts in short intervals converge to Poisson statistics. Lemke Oliver and Soundararajan [3] discovered biases in consecutive prime residues decaying as $O((\log\log x)/\log x)$; Holt [4] modeled these biases via sieve dynamics. However, no quantitative convergence rate for the overall gap distribution to Poisson has been established.

We construct a velocity field from the prime gap distribution and show it satisfies a nonlinear PDE whose unique stable equilibrium is the Poisson regime. This provides both a mechanism (the spectral gap of a Laguerre-type operator) and, numerically, a rate ($\approx (\log x)^{-0.19}$).

## 2. The Burgers–Fisher Equation for Prime Gaps

### 2.1. The Feller process and its Fokker–Planck equation.

Consider the Feller diffusion (Cox–Ingersoll–Ross process)

$$dV = (1-V)\,dt + \sqrt{2V}\,dW_t, \qquad V \geq 0, \tag{1}$$

with mean-reversion toward 1 and state-dependent noise $\sigma(V) = \sqrt{2V}$. Its Fokker–Planck equation for the density $f(v,t)$ is

$$\partial_t f = v f'' + (v+1) f' + f, \tag{2}$$

where primes denote $\partial_v$.

**Lemma 2.1.** *$f_\infty(v) = e^{-v}$ is the unique probability density satisfying $\mathcal{L} f_\infty = 0$, where $\mathcal{L} f = v f'' + (v+1) f' + f$.*

*Proof.* Direct verification: $\mathcal{L}(e^{-v}) = v \cdot e^{-v} + (v+1)(-e^{-v}) + e^{-v} = 0$. Uniqueness follows from the general theory of Feller diffusions [6]: the process (1) with $2 \times$ mean-reversion speed$/\sigma^2 = 2 \cdot 1/2 = 1 \geq 1$ has an entrance boundary at 0 and a natural boundary at $\infty$, admitting a unique stationary distribution. □

### 2.2. Derivation of the velocity equation.

**Theorem 2.2** (The sieve Burgers–Fisher equation)**.** *Let $f(v,t) > 0$ satisfy the Fokker–Planck equation* (2) *on* $(0,\infty)$*. Define the velocity field*

$$u(v,t) = -\frac{\partial}{\partial v} \log f(v,t) = -\frac{f'}{f}. \tag{3}$$

*Then $u$ satisfies the Burgers–Fisher equation in a porous medium:*

$$\boxed{\partial_t u + 2v\, u\, \partial_v u = v\, \partial_v^2 u + (v{+}2)\, \partial_v u + u(1{-}u).} \tag{4}$$

*Proof.* Set $\psi = \log f$, so $\psi' = f'/f$ and $f''/f = \psi'' + (\psi')^2$. From (2):

$$\psi_t = v(\psi'' + \psi'^2) + (v+1)\psi' + 1. \tag{5}$$

Differentiate (5) with respect to $v$:

$$\begin{aligned} \psi_t' &= \psi'' + v\psi''' + \psi'^2 + 2v\psi'\psi'' + \psi' + (v+1)\psi'' \\ &= v\psi''' + (v+2)\psi'' + 2v\psi'\psi'' + \psi'^2 + \psi'. \end{aligned}$$

Substitute $u = -\psi'$ (hence $\psi'' = -u'$, $\psi''' = -u''$):

$$-u_t = -vu'' - (v+2)u' + 2vu\,u' + u^2 - u.$$

Multiply by $-1$ and rearrange:

$$u_t = vu'' + (v+2)u' - 2vu\,u' - u^2 + u = vu'' + (v+2)u' + u(1-u) - 2vu\,u'. \quad \square$$

**Remark 2.3** (Structure of the equation)**.** Equation (4) combines three classical PDE mechanisms:

(1) *Burgers nonlinearity* $2vu\,u_v$: the hallmark of the Navier–Stokes and Burgers equations, arising here from the nonlinear transformation $u = -(\log f)'$.
(2) *Porous medium diffusion* $v\,u_{vv}$: the diffusivity equals $v$, so small gaps ($v \approx 0$) are "tight" and large gaps fluctuate freely.
(3) *Fisher–KPP reaction* $u(1-u)$: this term pushes $u$ toward 1 from either side. It emerges from the Fokker–Planck structure, not imposed by hand.

## 3. Spectral Stability of the Poisson Equilibrium

**Theorem 3.1** (Spectral analysis)**.**
[(i)]
(1) The unique bounded steady state of (4) is $u \equiv 1$, corresponding to the exponential distribution $f = e^{-v}$.
(2) The linearization around $u = 1$, writing $\delta u = u - 1$, is

$$\partial_t(\delta u) = v\,(\delta u)'' + (2-v)\,(\delta u)' - \delta u. \tag{6}$$

(3) The eigenvalue problem $v\,h'' + (2-v)\,h' + (\lambda+1)\,h = 0$ is the associated Laguerre equation with parameter $\alpha = 1$. Its solutions are the associated Laguerre polynomials $L_n^{(1)}(v)$ with eigenvalues

$$\lambda_n = -(n+1), \qquad n = 0, 1, 2, \ldots \tag{7}$$

(4) The spectral gap is $|\lambda_0| = 1$.

*Proof.* (i) A constant steady state $u = c$ satisfies $c(1-c) = 0$, giving $c = 0$ or $c = 1$. If $u = 0$ then $f' = 0$, i.e., $f = \text{const}$, which is not integrable on $[0, \infty)$. Thus $u = 1$ is the unique physically admissible steady state.

(ii) Substitute $u = 1 + \delta u$ into (4), keeping linear terms in $\delta u$:

$$\begin{aligned}(\delta u)_t + 2v \cdot 1 \cdot (\delta u)' &= v(\delta u)'' + (v+2)(\delta u)' + (1+\delta u)(1-1-\delta u)\\ &= v(\delta u)'' + (v+2)(\delta u)' - \delta u + O((\delta u)^2).\end{aligned}$$

Hence $(\delta u)_t = v(\delta u)'' + (2-v)(\delta u)' - \delta u$.

(iii) Setting $(\delta u)(v,t) = h(v)\,e^{\lambda t}$ gives $vh'' + (2-v)h' + (\lambda+1)h = 0$. With the substitution $\alpha = 1$ and $n = \lambda + 1$, this is the associated Laguerre equation $vh'' + (\alpha + 1 - v)h' + nh = 0$, whose polynomial solutions are $L_n^{(1)}(v)$ [5].

(iv) From $\lambda_n = -(n+1)$: $\lambda_0 = -1$, so all perturbation modes decay at rate $\geq 1$. $\square$

## 4. Convergence Under Decaying Sieve Forcing

**Theorem 4.1** (Perturbed convergence)**.** *Let $u(v,t)$ satisfy*

$$\partial_t u + 2vu\, u_v = vu_{vv} + (v+2)u_v + u(1-u) + \varepsilon(t)\, F(v), \tag{8}$$

*where $\|F\|_\infty \le M$ and $|\varepsilon(t)| \le C_0/t^\beta$ for $\beta > 0$. Then there exists $C > 0$ such that $\|\delta u(\cdot,t)\|_{L^2(v\,e^{-v})} \le C/t^\beta$ for all $t$ sufficiently large.*

*Proof.* Define the energy $E(t) = \frac{1}{2}\|\delta u\|^2_{L^2(ve^{-v})}$, where $L^2(ve^{-v})$ is the natural weighted space for the associated Laguerre polynomials.

**Step 1 (Dissipation).** By the spectral gap (Theorem 3.1(iv)): $\langle \mathcal{L}_{\text{lin}}(\delta u), \delta u\rangle \le -|\lambda_0| \cdot \|\delta u\|^2 = -\|\delta u\|^2$.

**Step 2 (Forcing).** $|\langle \varepsilon F, \delta u\rangle| \le |\varepsilon| \cdot M' \cdot \|\delta u\|$, where $M' = \|F\|_{L^2(ve^{-v})}$. By Young's inequality: $|\varepsilon| M' \|\delta u\| \le \frac{1}{2}\|\delta u\|^2 + \frac{M'^2}{2}\varepsilon^2$.

**Step 3 (Energy inequality).** $\dot{E} \le -2E + E + \frac{M'^2}{2}\varepsilon^2 = -E + \frac{C_1}{t^{2\beta}}$.

**Step 4 (Grönwall).** The integrating factor $e^t$ gives $E(t) \le E(t_0)e^{-(t-t_0)} + C_1 \int_{t_0}^t e^{-(t-\tau)}\tau^{-2\beta}\, d\tau$. The integral is $O(t^{-2\beta})$ by repeated integration by parts, yielding $E(t) = O(t^{-2\beta})$. □

## 5. Connection to Prime Gaps

**Hypothesis 5.1** (Sieve diffusion)**.** Under the Hardy–Littlewood conjecture, the normalized gap density $f(v,s)$ at scale $s = \log x$ can be written as $f = e^{-v}[1 + \varphi(v,s)]$, where the perturbation $\varphi$ is driven by the singular series and satisfies $\|\varphi(\cdot,s)\| = O(1/s^\beta)$ for some $\beta > 0$.

Under this hypothesis, the velocity field $u = 1 - \varphi'/(1+\varphi) \approx 1 - \varphi'$ satisfies a forced version of (4), and Theorem 4.1 gives $u \to 1$, i.e., convergence to Poisson.

**Remark 5.2** (What the hypothesis does not give)**.** Hypothesis 5.1 concerns the marginal gap density. The mutual information $\text{MI}(g_n; g_{n+1})$—a joint quantity—requires additional input. We verify the MI decay numerically (§6) but leave the rigorous joint implication as an open problem.

### 5.1. The sieve as porous medium.

Under the Hardy–Littlewood conjecture: $f(v,s) \propto W(vs) \cdot e^{-v}$, where $W(g) = \prod_{p|g,\, p>2}(p-1)/(p-2)$. Each prime $p$ is a "pore" with strength $\delta_p = 1/(p-2)$: $\delta_3 = 1$, $\delta_5 = 1/3$, $\delta_7 = 1/5$. As $s \to \infty$, gap values $g = vs$ grow and the mod-$p$ structure equidistributes, weakening each pore.

### 5.2. Exact computation: the mod-3 coupling.

**Proposition 5.3.** *Under i.i.d. uniform prime residues mod 3:* $\text{MI}(g_n \bmod 3;\, g_{n+1} \bmod 3) = 1/4$ *bit, with structural zeros at* $(1,1)$ *and* $(2,2)$ *in the joint table.*

*Proof.* With $r_n = p_n \bmod 3 \in \{1,2\}$ i.i.d. uniform, the gap $g_n \equiv r_{n+1} - r_n \pmod 3$. Enumerating all 8 triples $(r_n, r_{n+1}, r_{n+2}) \in \{1,2\}^3$ yields the joint distribution:

| $g_n \bmod 3 \setminus g_{n+1} \bmod 3$ | 0 | 1 | 2 |
|---|---|---|---|
| 0 | 2/8 | 1/8 | 1/8 |
| 1 | 1/8 | **0** | 1/8 |
| 2 | 1/8 | 1/8 | **0** |

Marginals: $P(0) = 1/2$, $P(1) = P(2) = 1/4$. Only entries $(1, 2)$ and $(2, 1)$ deviate from the product of marginals: $\mathrm{MI} = 2 \times \frac{1}{8} \log_2(\frac{1/8}{1/4 \cdot 1/4}) = \frac{1}{4}$ bit. □

## 6. Numerical Verification

All measurements use $N = 200{,}000$ primes at 11 scales ($10^3$ to $10^8$).

6.1. **MI decay rate.** $\mathrm{MI}_{\mathrm{total}}(x) \sim 0.555/(\log x)^{0.186}$, $R^2 = 0.986$. To our knowledge, this is the first numerical measurement of the MI decay rate for prime gaps.

6.2. **Prime 3 dominance.** $\mathrm{MI}_3 / \mathrm{MI}_{\mathrm{total}} \approx 94\%$ at all scales. Empirical $\mathrm{MI}_3 = 0.302$ bits vs. theoretical 0.250 bits; the 21% excess is the Lemke Oliver–Soundararajan bias [3], whose Main Conjecture predicts decay as $O((\log\log x)/\log x)$. The structural zeros $P(1, 1) = P(2, 2) = 0$ hold *exactly* for all primes $> 3$: if $g_n \equiv 1 \pmod 3$ then $p_{n+1} \equiv 2 \pmod 3$, forcing $g_{n+1} \equiv 1$ to require $p_{n+2} \equiv 0 \pmod 3$, which is impossible for $p_{n+2} > 3$. Verified empirically with zero exceptions in 78,493 consecutive gap pairs.

6.3. **Perturbation energy.** $E(s) = \|\varphi\|^2_{L^2(e^v)} \sim C/(\log x)^{0.63}$, $R^2 = 0.67$. The Laguerre decomposition shows the $n = 2$ mode (variance correction) dominates, while the $n = 1$ mode (mean shift) is nearly zero.

6.4. **Information cascade.**

| Prime $p$ | Decay exponent $\alpha_p$ | $R^2$ | MI at $10^8$ |
|---|---|---|---|
| 3 | 0.22 | 0.980 | 0.302 bits |
| 7 | 3.83 | 0.996 | 0.006 bits |
| 11 | 3.67 | 0.998 | 0.032 bits |

Prime 3 ($\alpha \approx 0.22$) is the bottleneck; primes $\geq 7$ decay as $(\log x)^{-3.5}$ or faster.

6.5. **Sieve diffusion model.** Starting from $f_0 = e^{-v}$ (Cramér) and iteratively adding sieve primes: adding $p = 3$ alone reduces KL divergence from the empirical distribution by 74% (from 0.061 to 0.016 nats at scale $10^7$).

## 7. Discussion: What the Paper Does and Does Not Prove

7.1. **The role of the PDE model.** Gallagher [1] already proved (conditionally) that prime gaps converge to Poisson. Our paper does not give a new proof of this fact. What the paper provides is a *mechanism*: a specific PDE (the Burgers–Fisher equation) whose spectral structure explains *why* the convergence occurs and *how fast* it proceeds. The relationship between our work and Gallagher's is analogous to the relationship between the Ising model and the thermodynamic observation that magnets lose magnetization above the Curie temperature: the observation was known first; the model explains the mechanism.

The logical structure is:

(1) **Pure PDE (Theorems 2.2–4.1):** If a density evolves under the Feller Fokker–Planck equation, the velocity field satisfies the Burgers–Fisher equation with stable equilibrium $u = 1$ and spectral gap 1. Perturbations with decaying forcing vanish. *These are unconditional theorems.*
(2) **Model (Hypothesis 5.1):** The prime gap density at scale $s = \log x$ is well-approximated by the Feller Fokker–Planck stationary distribution perturbed by the singular series, with perturbation decaying as $O(1/s^\beta)$. *This is a modeling assumption, supported by numerical evidence but not derived from number theory.*
(3) **Consequence:** Under the model, the spectral gap drives convergence to Poisson at a computable rate.

The open challenge is to *derive* Hypothesis 5.1 from the Hardy–Littlewood conjecture, which would close the gap between the PDE theorems and number theory.

### 7.2. Open problems.

(1) **Derive the sieve diffusion hypothesis** from Hardy–Littlewood and equidistribution of primes in arithmetic progressions.
(2) **Prove MI** $\to 0$ from the Burgers–Fisher framework (requires extending from marginal to joint gap densities).
(3) **Compute the exponent** $\alpha \approx 0.19$ from first principles, starting from the pore strength $\delta_3 = 1$ and the equidistribution rate of gaps mod 3.

## Computational Details

Python 3 with SymPy, NumPy, SciPy. Runtime $\approx 2$ minutes. Code, data, and interactive visualization: `https://github.com/tobiascanavesi/prime-fluid-dynamics`

**Use of generative AI.** Claude Opus 4.6 (Anthropic, 2025) assisted with code development and LaTeX formatting. All mathematical content is the original contribution of the author.

## References

[1] P. X. Gallagher, *On the distribution of primes in short intervals*, Mathematika **23** (1976), 4–9.
[2] G. H. Hardy and J. E. Littlewood, *Some problems of 'Partitio Numerorum': III*, Acta Math. **44** (1923), 1–70.
[3] R. J. Lemke Oliver and K. Soundararajan, *Unexpected biases in the distribution of consecutive primes*, PNAS **113** (2016), E4446–E4454.
[4] F. Holt, *Expected biases in the distribution of consecutive primes*, preprint (2024), arXiv:2405.03540.
[5] G. Szegő, *Orthogonal Polynomials*, 4th ed., AMS, 1975.
[6] H. Risken, *The Fokker–Planck Equation*, 2nd ed., Springer, 1989.

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